

Python-for-machine-learning / ST-1a.pdf

📄

⋮

Kiru0310 Add files via upload

983f98b · last week

🕒

154 KB

📄

✎

▼

3/13/26, 1:04 PM

Python-for-machine-learning/ST - 1(a).ipynb at main · Kiru0310/Python-for-machine-learning

Kiru0310 / Python-for-machine-learning

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

Python-for-machine-learning / ST - 1(a).ipynb

Kiru0310

Add files via upload

d9ede67 · 6 minutes ago

131 lines (131 loc) · 4.64 KB

PreviewCodeBlame

RawCopyDownloadEdit

Open in Colab

In []:

PROBLEM STATEMENT:
You are provided with the exam scores of 20 students in a class.
Write a python program to analyze these scores.Your analysis should include:
1.OVERALL CLASS PERFORMANCE: Calculate average,highest and lowest scores.
2.PASS/FAIL CLASSIFICATION:Identify which students passed the exam
3.GRADING ON A CURVE:Assign letter grades based on curved scoring or predefined thresholds

In [1]:

student_data = {
 "Alice": 85, "Bob": 42, "Charlie": 78, "David": 92, "Eve": 55,
 "Frank": 61, "Grace": 38, "Heidi": 89, "Ivan": 74, "Judy": 67,
 "Kevin": 50, "Liam": 49, "Mallory": 95, "Nina": 82, "Oscar": 70,
 "Peggy": 63, "Quentin": 58, "Riley": 45, "Sybil": 99, "Trent": 72
}

def analyze_class_data(data):
 scores = list(data.values())

 avg_score = sum(scores) / len(scores)
 high_score = max(scores)
 low_score = min(scores)

 print("--- CLASS SUMMARY ---")
 print(f"Average: {avg_score:.1f} | Highest: {high_score} | Lowest: {low_score}\n")

 print(f"{'Name':<10} | {'Score':<5} | {'Status':<6} | {'Grade':<5}")
 print("-" * 35)

 for name, score in data.items():

 status = "PASS" if score >= 50 else "FAIL"
 if score >= 90: grade = 'A'
 elif score >= 80: grade = 'B'
 elif score >= 70: grade = 'C'
 elif score >= 60: grade = 'D'
 else: grade = 'F'

 print(f"{name:<10} | {score:<5} | {status:<6} | {grade:<5}")

 analyze_class_data(student_data)

--- CLASS SUMMARY ---
Average: 68.2 | Highest: 99 | Lowest: 38

Name | Score | Status | Grade

Alice | 85 | PASS | B
Bob | 42 | FAIL | F
Charlie | 78 | PASS | C
David | 92 | PASS | A
Eve | 55 | PASS | F
Frank | 61 | PASS | D
Grace | 38 | FAIL | F
Heidi | 89 | PASS | B
Ivan | 74 | PASS | C
Judy | 67 | PASS | D
Kevin | 50 | PASS | F
Liam | 49 | FAIL | F
Mallory | 95 | PASS | A
Nina | 82 | PASS | B
Oscar | 70 | PASS | C
Peggy | 63 | PASS | D
Quentin | 58 | PASS | F
Riley | 45 | FAIL | F
Sybil | 99 | PASS | A
Trent | 72 | PASS | C

